

Vector form of a system of linear equations

Definition

Consider the SLEs

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

Such a system can be expressed in **vector form** or as a **vector equation** by using **linear combinations** of column vectors:

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Problem

Express the following system of linear equations in vector form.

$$\begin{array}{ccccccccc} 2x_1 & + & 4x_2 & - & 3x_3 & = & -6 \\ & & x_2 & + & 5x_3 & = & 0 \\ x_1 & + & x_2 & + & 4x_3 & = & 1 \end{array}$$

$$x_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 1 \end{bmatrix}$$

Definition

Let $A = [a_{ij}]$ be an $m \times n$ matrix with columns $A_1, A_2, \dots, A_n$, written $A = [A_1 \ A_2 \ \dots \ A_n]$, and let X be an $n \times 1$ column vector,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Then the product of matrix A and (column) vector X is the $m \times 1$ column vector given by

$$\begin{bmatrix} A_1 & A_2 & \dots & A_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 A_1 + x_2 A_2 + \dots + x_n A_n = \sum_{j=1}^n x_j A_j$$

that is, AX is a linear combination of the columns of A .

Problem

Express the following system of linear equations in matrix form.

$$\begin{array}{rrcrcl} 2x_1 & + & 4x_2 & - & 3x_3 & = & -6 \\ & & x_2 & + & 5x_3 & = & 0 \\ x_1 & + & x_2 & + & 4x_3 & = & 1 \end{array}$$

$$\begin{bmatrix} 2 & 4 & -3 \\ 0 & -1 & 5 \\ 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 1 \end{bmatrix}$$

$$A \vec{X} = x_1 \vec{A}_1 + x_2 \vec{A}_2 + x_3 \vec{A}_3 = \vec{b}$$

$$\vec{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$A = \begin{bmatrix} \vec{A}_1 & \vec{A}_2 & \vec{A}_3 \end{bmatrix}$$

Problem

Find all values of a for which the system

$$\begin{array}{ccccccccc} x & + & y & & & = & 0 \\ & & ay & + & z & = & 0 \\ x & + & y & + & az & = & 0 \end{array}$$

has nontrivial solutions, and determine the solutions.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 1 & 1 & a & 0 \end{array} \right] \xrightarrow{-r_1 + r_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 0 & 0 & a & 0 \end{array} \right]$$

Case (I): $a \neq 0$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 0 & 0 & a & 0 \end{array} \right] \rightarrow z=0, y=0, x=0 \text{ which is the trivial solution and hence when } a \neq 0 \text{ there is no nontrivial solution.}$$

Case (II): $a = 0$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \boxed{\begin{array}{l} x = -a, \\ y = a, \\ z = 0 \\ s \in \mathbb{R} \end{array}} \rightarrow X = a_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \text{ where } a_1 \in \mathbb{R}$$

Theorem

Suppose the matrix equation $A\vec{x} = \vec{b}$ is consistent and let $\vec{p} \neq \vec{0}$ be a solution. Assume that $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$, then the solution set of $A\vec{x} = \vec{b}$ is the set of all the vectors of the form $\vec{w} = \vec{p} + \vec{v}_h$.

Problem

Solve the following systems and explain the relation between their solution sets.

$$\begin{array}{rrrrr} x & + & y & + & z & = & 0 \\ x & + & y & - & z & = & 0 \\ x & + & y & & & = & 0 \end{array}$$

$$\begin{array}{rrrrr} x & + & y & + & z & = & -1 \\ x & + & y & - & z & = & 1 \\ x & + & y & & & = & 0 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left. \begin{array}{l} x = -s \\ y = s \\ z = 0 \end{array} \right\} \rightarrow \vec{v}_h = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad s \in \mathbb{R}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & -1 & 1 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left. \begin{array}{l} x = -s \\ y = s \\ z = -1 \end{array} \right\} \rightarrow \vec{w} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

Note that $\vec{p} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$ is a solution of the non-homogeneous SLE

whereas $s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \vec{v}_h$ is the solution of the homogeneous SLE.

$\vec{p}$ is a vector in $\mathbb{R}^3$, whereas $s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ corresponds to a line in $\mathbb{R}^3$.

See the Geogebra link in Camino.

Span of a Set of Vectors

Definition

Let $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k\}$ be a set of vectors in $\mathbb{R}^n$. Then the collection of all linear combinations of these vectors is called the **span** of these vectors, written $\text{Span}\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k\}$.

Problem

Describe the span of the vectors $\vec{u} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$.

Note that any linear combination of $\vec{u}$ and $\vec{v}$ is of the form $a_1\vec{u} + a_2\vec{v} = a_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ a_1 + 2a_2 \\ 2a_1 + 3a_2 \end{bmatrix}$ which can be thought of as a vector in the yz -plane.

Also, consider an arbitrary vector $\begin{bmatrix} 0 \\ y \\ z \end{bmatrix}$ in the yz -plane. We can write any such vector as a linear combination of $\vec{u}$ and $\vec{v}$:

$$\begin{bmatrix} 0 \\ y \\ z \end{bmatrix} = x_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = (-3y + 2z) \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + (2y - z) \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 2 & y \\ 2 & 3 & z \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & y \\ 0 & -1 & z - 2y \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -3y + 2z \\ 0 & 1 & 2y - z \end{array} \right] \rightarrow \begin{cases} x_1 = -3y + 2z \\ x_2 = 2y - z \end{cases}$$

The $\text{Span}\{\vec{u}, \vec{v}\}$ is the yz -plane.

Span of a Set of Vectors

Question

Consider the previous example where the span of $\vec{u}$ and $\vec{v}$ was the YZ -plane. Suppose we add another vector $\vec{w}$, and consider the span of $\vec{u}$, $\vec{v}$, and $\vec{w}$. What would happen to the span?

Case (I): Suppose $\vec{w}$ is also a vector in the yz -plane. For example, consider $\vec{w} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, then $\vec{w}$ is already in the $\text{Span}\{\vec{u}, \vec{v}\}$.

$$\vec{w} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = -5\vec{u} + 3\vec{v} = -5 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

$$\text{Hence, } \text{Span}\{\vec{u}, \vec{v}, \vec{w}\} = \text{Span}\{\vec{u}, \vec{v}\}$$

Case (II): Suppose $\vec{w}$ is not in the yz -plane. For example, $\vec{w} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Notice that any vector in $\mathbb{R}^3$ can be written as a linear combination of $\vec{u}$, $\vec{v}$, and $\vec{w}$ (see below). Hence, $\text{Span}\{\vec{u}, \vec{v}, \vec{w}\} = \mathbb{R}^3$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x_1 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

where $x_1 = 2z - 3y - 3x$
 $x_2 = 2x + 2y - z$
 $x_3 = x$

$$\begin{aligned} & \left[\begin{array}{ccc|c} 1 & 2 & -1 & y \\ 2 & 3 & 0 & z \\ 0 & 0 & 1 & x \end{array} \right] \xrightarrow{-2r_3 + r_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & y \\ 0 & -1 & 2 & z - 2y \\ 0 & 0 & 1 & x \end{array} \right] \xrightarrow{-2r_3 + r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2z - 3y - 3x \\ 0 & -1 & 0 & z - 2y - 2x \\ 0 & 0 & 1 & x \end{array} \right] \xrightarrow{r_3 + r_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2z - 3y - 3x \\ 0 & -1 & 0 & z - 2y - 2x \\ 0 & 0 & 1 & x \end{array} \right] \xrightarrow{-r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2z - 3y - 3x \\ 0 & 1 & 0 & 2x + 2y - z \\ 0 & 0 & 1 & x \end{array} \right] \end{aligned}$$

Linearly Independent Set of Vectors

Definition

Let $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k\}$ be a set of vectors in $\mathbb{R}^n$. This set is **linearly independent** if no vector in the set is in the span of the other vectors of that set.

Problem

Consider the vectors

$$\vec{u} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \vec{v} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}, \vec{w} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Is the set $\{\vec{u}, \vec{v}, \vec{w}\}$ linearly independent?

(I) We can easily see that $\vec{w}$ is not in $\text{Span}\{\vec{u}, \vec{v}\}$.

$$\text{(II)} \quad x_1 \vec{v} + x_2 \vec{w} = \vec{u} \Rightarrow \begin{bmatrix} 0 & 1 & | & 0 \\ 2 & 0 & | & 1 \\ 3 & 2 & | & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{1}{2} \\ 3 & 2 & | & 2 \\ 0 & 1 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{1}{2} \\ 0 & 2 & | & \frac{1}{2} \\ 0 & 1 & | & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{1}{2} \\ 0 & 1 & | & \frac{1}{4} \\ 0 & 1 & | & 0 \end{bmatrix} \\ \rightarrow \begin{bmatrix} 1 & 0 & | & \frac{1}{2} \\ 0 & 1 & | & \frac{1}{4} \\ 0 & 0 & | & -\frac{1}{4} \end{bmatrix} \rightarrow \text{no solution} \rightarrow \vec{u} \text{ is not in } \text{Span}\{\vec{v}, \vec{w}\}$$

(III) Similarly, one can show that $x_1 \vec{u} + x_2 \vec{w} = \vec{v}$ has no solution which means $\vec{v}$ is not in $\text{Span}\{\vec{u}, \vec{w}\}$. From (I), (II), and (III), we can conclude that $\{\vec{u}, \vec{v}, \vec{w}\}$ is a linearly independent set.